RELATIVE APPROXIMATION OF FORMAL LOCAL MAPS BEFORE ETALE LOCALISATION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 11143-11203 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 11205-11283, the complete author proof (77 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Local maps of completions, finite type modifications, and the control lemmas for approximation; the reduction is retained.

Source reference. Original lemma lemma-isomorphism-approximation: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11071–11095. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-slice-smooth: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10775–10783. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma limits-lemma-descend-etale: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11283–11283. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma limits-lemma-descend-finite-presentation: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11283–11283. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma limits-proposition-characterize-locally-finite-presentation: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11283–11283. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-algebra-proposition-Noetherian-complete-G-ring: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11283–11283. Cited by the retained statement or proof but not part of the collected spans;

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma smoothing-theorem-popescu: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11283–11283. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $X \to S$, $Y \to T$, x, s, y, t, σ, y_σ, and φ be given as follows: we have morphisms of schemes*

$$\begin{array}{ccc} X & Y & \\ \downarrow & \downarrow & \text{with points} \\ S & T & \end{array} \qquad \begin{array}{cc} x & y \\ \downarrow & \downarrow \\ s & t \end{array}$$

Here S is locally Noetherian and T is of finite type over $\mathbf{Z}$. The morphisms $X \to S$ and $Y \to T$ are locally of finite type. The local ring $\mathcal{O}_{S,s}$ is a G-ring. The map

$$\sigma : \mathcal{O}_{T,t} \longrightarrow \mathcal{O}^\wedge_{S,s}$$

is a local homomorphism. Set $Y_\sigma = Y \times_{T,\sigma} \mathrm{Spec}(\mathcal{O}^\wedge_{S,s})$. Next, y_σ is a point of Y_σ mapping to y and the closed point of $\mathrm{Spec}(\mathcal{O}^\wedge_{S,s})$. Finally

$$\varphi : \mathcal{O}^\wedge_{X,x} \longrightarrow \mathcal{O}^\wedge_{Y_\sigma, y_\sigma}$$

is an isomorphism of $\mathcal{O}^\wedge_{S,s}$-algebras. In this situation there exists a commutative diagram

$$\begin{array}{ccccccc} X & \longleftarrow & W & \longrightarrow & Y \times_{T,\tau} V & \longrightarrow & Y \\ \downarrow & & & \searrow \quad \swarrow & & & \downarrow \\ S & \longleftarrow & & V & \xrightarrow{\tau} & & T \end{array}$$

of schemes and points $w \in W$, $v \in V$ such that

(1) *$(V, v) \to (S, s)$ is an elementary étale neighbourhood,*
(2) *$(W, w) \to (X, x)$ is an elementary étale neighbourhood, and*
(3) *$\tau(v) = t$.*

Let $y_\tau \in Y \times_T V$ correspond to y_σ via the identification $(Y_\sigma)_s = (Y \times_T V)_v$. Then

(4) *$(W, w) \to (Y \times_{T,\tau} V, y_\tau)$ is an elementary étale neighbourhood.*

Proof. Denote $X_\sigma = X \times_S \mathrm{Spec}(\mathcal{O}^\wedge_{S,s})$ and $x_\sigma \in X_\sigma$ the unique point lying over x. Observe that $\mathcal{O}^\wedge_{S,s}$ is a G-ring by More on Algebra, Proposition more-algebra-proposition-Noetherian-complete-G-ring. By Lemma lemma-isomorphism-approximation we can choose

$$(X_\sigma, x_\sigma) \leftarrow (U, u) \to (Y_\sigma, y_\sigma)$$

where both arrows are elementary étale neighbourhoods.

After replacing S by an open neighbourhood of s, we may assume $S = \mathrm{Spec}(R)$ is affine. Since $\mathcal{O}_{S,s}$ is a G-ring by Smoothing Ring Maps, Theorem smoothing-theorem-popescu the ring $\mathcal{O}^\wedge_{S,s}$ is a filtered colimit of smooth R-algebras. Thus we can write

$$\mathrm{Spec}(\mathcal{O}^\wedge_{S,s}) = \lim S_i$$

as a directed limit of affine schemes S_i smooth over S. Denote $s_i \in S_i$ the image of the closed point of $\mathrm{Spec}(\mathcal{O}^\wedge_{S,s})$. Observe that $\kappa(s) = \kappa(s_i)$. Set $X_i = X \times_S S_i$ and denote $x_i \in X_i$ the unique point mapping to x. Note that $\kappa(x) = \kappa(x_i)$. Since T is of finite type over $\mathbf{Z}$ by Limits, Proposition limits-proposition-characterize-locally-finite-presentation we can choose an i and a morphism $\sigma_i : (S_i, s_i) \to (T, t)$ of pointed schemes whose composition with $\mathrm{Spec}(\mathcal{O}^\wedge_{S,s}) \to S_i$ is equal to σ. Set

$Y_i = Y \times_T S_i$ and denote y_i the image of y_σ. Note that $\kappa(y_i) = \kappa(y_\sigma)$. By Limits, Lemma limits-lemma-descend-finite-presentation we can choose an i and a diagram

$$\begin{array}{ccccc} X_i & \longleftarrow & U_i & \longrightarrow & Y_i \\ & \searrow & \downarrow & \swarrow & \\ & & S_i & & \end{array}$$

whose base change to $\mathrm{Spec}(\mathcal{O}^\wedge_{S,s})$ recovers $X_\sigma \leftarrow U \rightarrow Y_\sigma$. By Limits, Lemma limits-lemma-descend-etale after increasing i we may assume the morphisms $X_i \leftarrow U_i \rightarrow Y_i$ are étale. Let $u_i \in U_i$ be the image of u. Then $u_i \mapsto x_i$ hence $\kappa(x) = \kappa(x_\sigma) = \kappa(u) \supset \kappa(u_i) \supset \kappa(x_i) = \kappa(x)$ and we see that $\kappa(u_i) = \kappa(x_i)$. Hence $(X_i, x_i) \leftarrow (U_i, u_i)$ is an elementary étale neighbourhood. Since also $\kappa(y_i) = \kappa(y_\sigma) = \kappa(u)$ we see that also $(U_i, u_i) \rightarrow (Y_i, y_i)$ is an elementary étale neighbourhood.

At this point we have constructed a diagram

$$\begin{array}{ccccccccc} X & \longleftarrow & X \times_S S_i & \longleftarrow & U_i & \longrightarrow & Y \times_T S_i & \longrightarrow & Y \\ \downarrow & & & \searrow & \downarrow & \swarrow & & & \downarrow \\ S & \longleftarrow & & & S_i & & \longrightarrow & & T \end{array}$$

as in the statement of the lemma, except that $S_i \rightarrow S$ is smooth. By Lemma lemma-slice-smooth and after shrinking S_i we can assume there exists a closed subscheme $V \subset S_i$ passing through s_i such that $V \rightarrow S$ is étale. Setting W equal to the scheme theoretic inverse image of V in U_i we conclude. $\square$